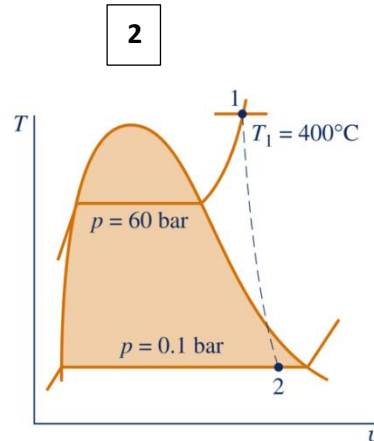
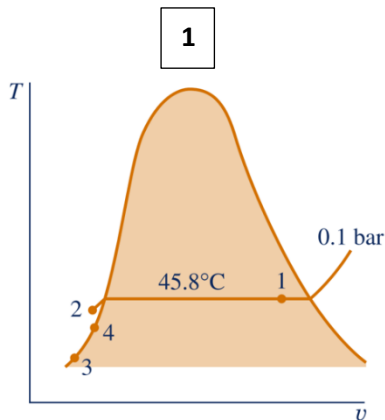


Name:

11/10/2021

Question 1. (2 points)

Each of the diagrams shows the processes that result from adding a device to a control volume. Determine the added device in each control volume and describe the process.



1) Condenser

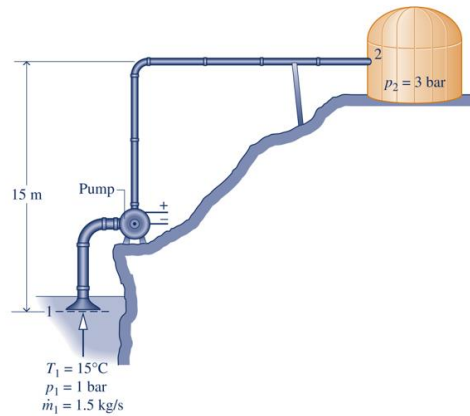
Because of cooling system in condenser, both phases will be occurred in first inlet so the state is saturated (L.V) . In outlet (state 2) liquid is expected. States 3 and 4 only saturated liquid is presented as the Substance for cooling system.

2) Turbine

In inlet vapor comes into the system and in outlet because of turbine blades and decreasing in temperature, some of vapor converted to liquid so we have saturated (L.V) in outlet.

Question 2. (3 points)

Figure provides steady-state operating data for a pump drawing water from a reservoir and delivering it at a pressure of 3 bar to a storage tank perched above the reservoir. The mass flow rate of the water is 1.5 kg/s. The water temperature remains nearly constant at 15 °C, there is no significant change in kinetic energy from inlet to exit, and heat transfer between the pump and its surroundings is negligible. Determine the power required by the pump, in kW. Let $g = 9.81 \text{ m/s}^2$



$$\dot{W} = \dot{m}((h_1 - h_2) + g(z_1 - z_2))$$

$$h_f(T_1) = 62.982 \frac{\text{kJ}}{\text{kg}}; v(T_1) = 0.001001 \frac{\text{m}^3}{\text{kg}}; P_{\text{sat}}(T_1) = 1.7057 \text{ kPa}; P_1 = 100 \text{ kPa}; P_2 = 300 \text{ kPa}$$

$$h_1 = h_f(T_1) + v_f(T_1) * [P_1 - P_{\text{sat}}(T_1)] = 62.982 + 0.001001 * [100 - 1.7057] = 63.08039259$$

$$h_2 = h_f(T_2) + v_f(T_2) * [P_2 - P_{\text{sat}}(T_2)] = 62.982 + 0.001001 * [300 - 1.7057] = 63.28059259$$

$$\dot{W} = 1.5 \left((63.08039259 - 63.28059259) + 9.81 * (-15) * \frac{1}{1000} \right) = -0.521025 \text{ kW}$$